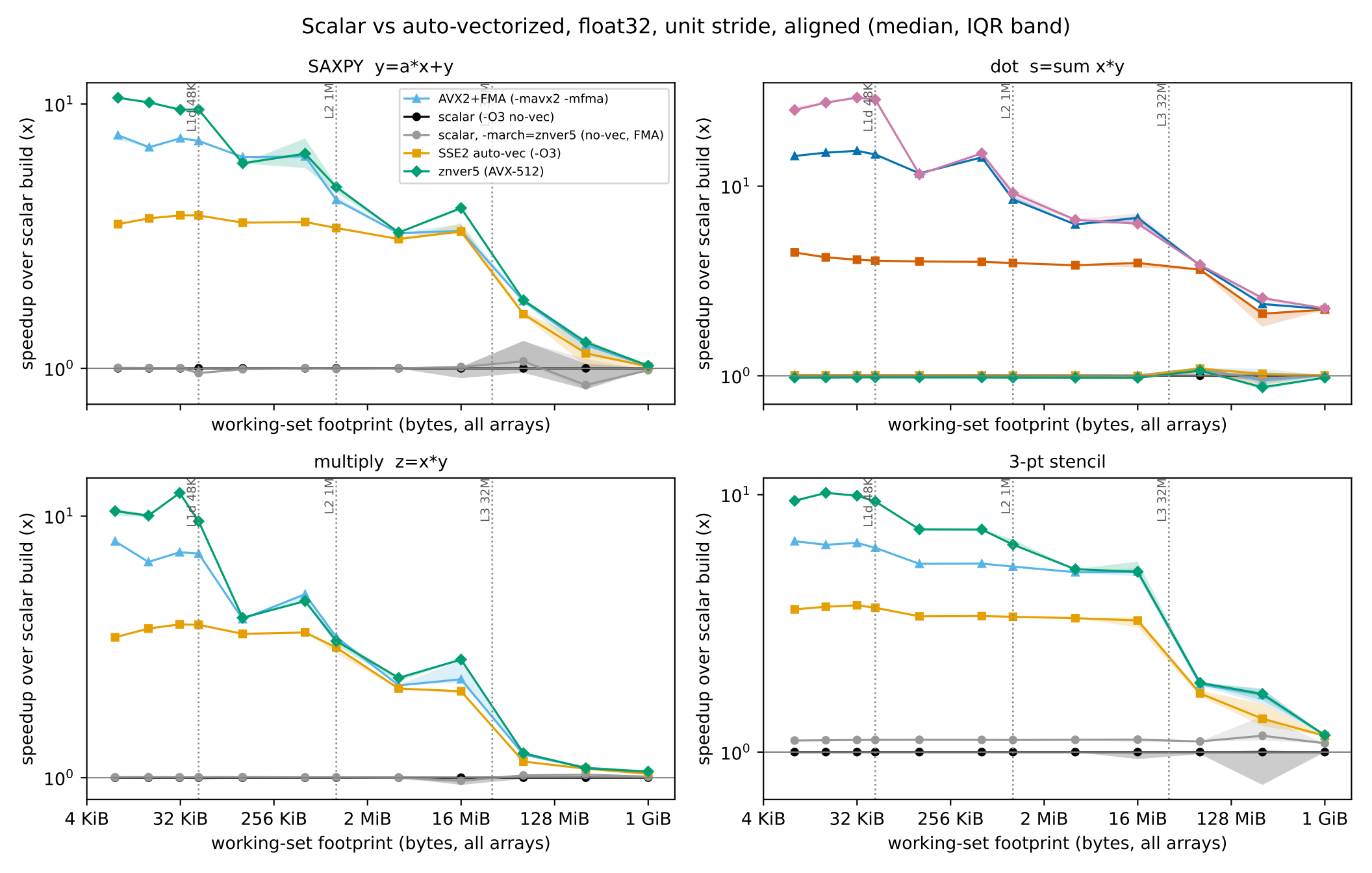
Scalar vs auto-vectorized, float32, unit stride, aligned (median, IQR band)
SAXPY y=a*x+y
dot s=sum x*y
multiply z=x*y
3-pt stencil
AVX2+FMA (-mavx2 -mfma)
scalar (-O3 no-vec)
scalar, -march=znver5 (no-vec, FMA)
SSE2 auto-vec (-O3)
znver5 (AVX-512)
L1d 48K
L2 1M
L3 32M
speedup over scalar build (x)
working-set footprint (bytes, all arrays)
4 KiB
32 KiB
256 KiB
2 MiB
16 MiB
128 MiB
1 GiB